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The influence of CAD model continuity on accuracy and productivity of CNC machining

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Abstract

Efficient and productive manufacturing of freeform shapes requires a suitable three-dimensional CAD model at the entrance to the CAM system. The paper deals with the impact of NURBS or B-spline CAD model geometric continuity on the accuracy and productivity of 5-axis ball-end milling of freeform surfaces. The relationships between a different order of CAD model geometric continuity and the quality of the toolpath generated in CAM system is analysed and demonstrated on example of Blisk blade profile. In order to reveal the pure effect of CAD geometry on the quality of the machined surface, linear interpolation of individual positions of CL points is considered and no further smoothing of the toolpath is applied. In addition to the commonly used indicator of toolpath quality represented by the distance of CL points, the discrete curvature of a toolpath in a form of a piecewise linear discrete curve is introduced and its dependence on the curvature and continuity of the underlying CAD model is demonstrated. It is shown that the higher order of CAD model geometric continuity significantly eliminates sharp changes in the distance of CL points and smoothes the discrete curvature of the toolpath. Finally, it is experimentally verified that increasing the continuity of the CAD model from $\boldsymbol{G^0}$ to $\boldsymbol{G^3}$, while maintaining the same cutting conditions, leads to

an increase in workpiece accuracy and a reduction in machining time, without the need to smooth the toolpath generated in the CAM system.

Keywords: CAD model, discrete curve curvature, geometric continuity, surface accuracy, 5-axis milling, toolpath smoothness, machining productivity, Blisk machining

1 Introduction

In 5-axis ball-end milling of freeform surfaces, the emphasis today is on constantly reducing production costs while increasing the quality and accuracy of the final workpiece. This trend is especially evident in complex components in the aerospace and energy industries, such as compressors or turbine blades. In both cases, it is an effort to increase the efficiency of turbomachines by using, for example, new difficult-to-cut materials, more complex shapes and thinner blades. The production of paddles usually consists of several stages: roughing, semifinishing and finishing. Due to the complexity of the new blade shapes, it is necessary to finish the surfaces by spot machining, most often by a 5-axis interlacing strategy. For this, milling tools with a circular cutting edge are used. Achieving the required accuracy and quality of surfaces is now solved mainly by the choice of toolpath density and suitable cutting conditions. However, this is not always sufficient, and even with the right choice, the quality and accuracy of the surface often deteriorates. Preparing these complex workpieces for the milling on CNC (Computer Numerical Control) machines is a very complex and interdisciplinary process in which it is necessary to pay attention to the quality of data flow in the entire chain from part design to machining on a CNC machine. Any shortcoming in the preparation of production can mean deterioration in productivity, or production of a poor-quality workpiece, which results either scrap or needs other time-consuming post-processing adjustments.

The main levels in the chain from part design in CAD (Computer Aided Design) system, processing in CAM (Computer Aided Manufacturing) system to production and inspection can be represented in blocks as

CAD model – CAM system – Postprocessor – CNC control system – Machine tool – Machining process – Measuring.

Many researches focus on milling optimisations in the CAM level [1, 2], but this may not be enough, as the entire manufacturing process begins in a 3-dimensional CAD model. CAD-level errors can therefore be carried on to the machining process itself and thus adversely affect the result. For this reason, our research focuses on quality analysis at the first level, i.e. the phase of CAD modelling and examination of the impact of a CAD model quality on all subsequent levels of 5-axis milling.

One of the requirements in the preparation of 5-axis milling is to achieve constant cutting conditions. The first task is to achieve the smoothest possible feedrate of the cutting tool over the machined free-form surface. Since the 5-axis toolpath of the CAM is composed of linear interpolations, which introduces discontinuity into the CNC processing, jumps of velocity and acceleration, and jerk can occur in the toolpath. This can lead, for example, to the risk of vibration [3] or deterioration of surface roughness [4].

An improvement in the smoothness of the feedrate can be achieved by toolpath spline compensation with respect to the kinematic parameters of the motion axes drives of the given machine tool, [5–7]. Due to the fluctuating feedrate, the cutting force load between the tool and the workpiece also changes – in particular, by uneven material removal rate, and thus, by uneven feed per tooth. This can lead to tool deflection and reduced accuracy. The response to tool deflection is to correct the original toolpath so that the resulting deviations after machining are minimized [8, 9]. In the case where the part has a similar flexibility as the cutting tool, it is necessary to solve the deflection of the part itself due to the cutting forces [10].

Due to the flexibility of the part and the cutting tool in combination with the excitation of the cutting process, forced vibrations or chatter can occur. A useful intervention is to control variable spindle speed on the basis of simulations so as to avoid problematic frequencies at the given point in the toolpath [11, 12]. Spindle speed control can also be applied to maintain a constant cutting speed at the point of contact between the tool and the workpiece. This makes it possible to increase productivity while reducing surface roughness. Optimisation takes place on the CAM toolpath of the tool [13]. Further optimisation reduces fluctuations of cutting speed during machining due to the intervention in the CAM toolpath and the change of the tilt and lead angles [14]. The influence of the choice of the tilt and lead angles has a direct effect on the accuracy, surface quality [15], roughness and stability of machining [16–18].

All the above mentioned approaches are focused on improvements in the technological phase, i.e. smoothing the toolpath and optimal setting of cutting conditions. However, the input data for all subsequent technological processes are derived from the CAD model of the manufactured workpiece. Obviously, the high quality of geometrical properties of CAD model subsequently results in increased efficiency of the machining process and improved quality of the machined surface. Therefore, in this paper, the influence of geometrical quality and properties of CAD model on the quality and efficiency of CNC machining is examined and demonstrated on an example inspired by turbine blade manufacturing.

The paper is organised as follows. In Section 2, the basic mathematical preliminaries from computer geometry are briefly mentioned. Suitable metrics to investigate, evaluate and assess the influence of various CAD geometries on quality of toolpath are introduced in Section 3. Section 4 shows a motivational example where the simulation, analysis and comparison of a toolpath for 5-axis ball-end milling generated from a CAD model with G^1 geometrical continuity

commonly found in technical practice and a CAD models with higher order of geometric continuity are given. A practical experiment taking into account the conclusions from the motivational example is described in Section 5. Section 6 concludes the paper.

2 Mathematical-geometrical preliminaries

Geometrical properties and quality of CAD model are significantly influenced by mathematical representation on which the CAD model is based. The mathematical representation implemented in CAD systems is usually based on NURBS (Non-Uniform Rational B-Spline) representation [19]. This representation allows not only a comprehensive description of the shape of the modelled object and its simple modification [20], but also provides a tool for analysing the shape and its geometric properties [21]. However, NURBS, as well as its special types B-spline and Bézier, represent still a relatively mysterious theory not fully understood by users of CAD and CAM systems. Consequently, interactive modelling of freeform shapes with the required geometric properties is moved to an intuitive level, even if the effects of modification of the shape by means of available tools and mechanisms are often unpredictable [22].

In this section, the most important definition and geometrical properties of NURBS and B-spline curves are briefly summarised, including rules for analysis of geometric continuity by means of curvature graph. Since the toolpath generated in the CAM system is a piecewise linear curve, discrete curvature is introduced here as a metric to evaluate the smoothness of the toolpath.

2.1 NURBS, B-spline and Bézier curve

NURBS curve $\mathbf{C}(u) = (x(u), y(u), z(u))$ is a piecewise defined curve approximating control points $\mathbf{P}_i$, $i = 0, ..., n$, with rational parameterisation given by vector equation [19]

$$\mathbf{C}(u) = \frac{\sum_{i=0}^{n} w_i N_{i,p}(u)\mathbf{P}_i}{\sum_{i=0}^{n} w_i N_{i,p}(u)}, \ u \in [u_p, u_{n+1}], \tag{1}$$

where $\mathbf{C}(u)$ is a vector value function of one variable u, w_i, $i = 0, ..., n$, are weights of control points, $N_{i,p}(u)$ are piecewise B-spline basis functions of p-th degree defined on a non-uniform knot vector

$$U = (u_0, ..., u_{n+p+1}), \ u_i \leq u_{i+1}, \tag{2}$$

calculated according to the recurrent formula

$$\begin{aligned} N_{i,0}(u) &= \begin{cases} 1 & u \in [u_i, u_{i+1}) \\ 0 & \text{otherwise} \end{cases} \\ N_{i,p}(u) &= \tfrac{u-u_i}{u_{i+p}-u_i} N_{i,p-1}(u) + \tfrac{u_{i+p+1}-u}{u_{i+p+1}-u_{i+p}} N_{i+p,p-1}(u), \end{aligned} \tag{3}$$

where $\frac{0}{0} = 0$ is defined.

B-spline curve is a special case of NURBS curve with the same weights of all control points (usually $w_i = 1$, $i = 0, ..., n$).

2.2 Parametric and geometric continuity of curves

Two differentiable curves $\mathbf{C}(u)$, $u \in [0, 1]$, and $\mathbf{K}(v)$, $v \in [0, 1]$, are C^K continuous at common point $\mathbf{C}(1) = \mathbf{K}(0)$ if the first K parametric derivatives match at the common point [23], i.e.

$$\mathbf{C}^{(i)}(1) = \mathbf{K}^{(i)}(0), \ i = 1, .., K. \tag{4}$$

C^K continuity (4) of two curves is referred to as the *parametric continuity* of K-th order [19].

To ensure and analyse parametric continuity (4), it is necessary to know the specific mathematical expression of the curves. For practical applications in CAD/CAM systems, a parameterisation independent measure of continuity G^K called *geometric continuity* of K-th order is introduced in the following way [24, 25]: curves $\mathbf{C}(u)$ and $\mathbf{K}(v)$ are G^K continuously connected at the common point $\mathbf{C}(1) = \mathbf{K}(0)$ if there exists a parameterisation $\bar{\mathbf{C}}(u)$ equivalent to $\mathbf{C}(u)$ such that $\bar{\mathbf{C}}(u)$ and $\mathbf{K}(v)$ meet with C^K continuity at the common point. It follows that if two curves are connected with C^K continuity, they are connected with G^K continuity, too.

Geometric continuity is defined on the basis of geometrical invariants such as unit tangent vector

$$\hat{\mathbf{t}}(u) = \frac{\mathbf{C}'(u)}{\|\mathbf{C}'(u)\|} \tag{5}$$

and curvature vector [26]

$$\mathbf{k}(u) = \frac{\mathbf{C}'(u) \times \mathbf{C}''(u) \times \mathbf{C}'(u)}{\|\mathbf{C}'(u)\|^4}. \tag{6}$$

Tangent vector (5) and curvature vector (6) are perpendicular. The length of curvature vector (6) is called the first curvature $k(u)$ of the curve.

Geometrical invariants of a curve are related to arc-length parameterisation [27]. The arc length $s(u)$ of a regular curve $\mathbf{C}(u)$, $u \in I$, from point $\alpha \in I$ is given by

$$s(u) = \int_{\alpha}^{u} \|\mathbf{C}'(u)\| \, \mathrm{d}u, \tag{7}$$

where $\|\mathbf{C}'(u)\|$ is the length of tangent vector. In arc-length parameterisation, where the parameter u is arc length (7) measured from point α, the length of tangent vector is equal to one for all u, i.e.

$$\frac{\mathrm{d}s(u)}{\mathrm{d}u} = 1 = \|\mathbf{C}'(u)\| \, . \tag{8}$$

Consequently, the curve parameterised by arc length s has unit length of tangent vector $\|\mathbf{C}'(s)\| = 1$ and the curvature vector $\mathbf{k}(s)$ is given by the second derivative

$$\mathbf{k}(s) = \mathbf{C}''(s). \tag{9}$$

A visualisation of curvature vector (6) implemented in CAD/CAM systems is known as a curvature graph. Usually, it is possible to set a suitable density and scale of the curvature graph. Curvature graph represents a robust tool not only for assessing the shape of a single curve but also for geometric continuity up to third order of two curves analysis, see fig. 1.

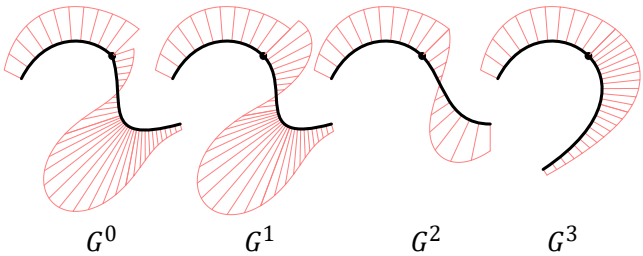


Fig. 1 Continuity analysis by means of curvature graph

The curvature graph provides the following information for the geometric continuity analysis:

- G^0 – position continuity measures location of endpoints of two curves

$$\mathbf{C}(1) = \mathbf{K}(0), \tag{10}$$

  i.e. $G^0 = C^0$. If two curves are G^0 continuously joined, the curvature graphs have different directions at the common point.
- G^1 – tangency continuity measures position and curve direction at the endpoints of two curves. If two curves are G^1 continuously joined, the unit tangent vectors (tangent lines) are the same at the common point

$$\hat{\mathbf{t}}_{\mathbf{C}}(1) = \hat{\mathbf{t}}_{\mathbf{K}}(0). \tag{11}$$

  Consequently, the curvature graphs have the same direction at the common point of the two curves but it is not continuous.
- G^2 – curvature continuity measures position, curve direction and curvature at the endpoints of two curves. If two curves are G^2 continuously joined, the curvature vector is continuous

$$\mathbf{k}_{\mathbf{C}}(1) = \mathbf{k}_{\mathbf{K}}(0), \tag{12}$$

  i.e. the curvature graphs are continuous (but not smooth) at the common point of the two curves.

- G^3 – geometric continuity of third order measures position, curve direction, curvature and its derivative at the endpoints of two curves. If two curves are G^3 continuously joined, the first derivatives of curvature vectors are continuous

$$\mathbf{k'_C}(1) = \mathbf{k'_K}(0), \tag{13}$$

  i.e. the curvature graphs are smooth at the common point of the two curves.

As mentioned above, if two curves are C^K continuously joined, they are G^K continuously joined, too. However, the opposite implication does not apply, i.e. G^K continuity does not mean that the derivatives of the curves are equal at the common point up to K-th order (as stated in manual [28], for example).

3 Metrics to assess toolpath quality

The quality and efficiency of the manufacturing process is highly influenced by the behaviour of a toolpath. Sharp changes of the toolpath result in uneven machining conditions and quality deterioration. In order to reveal the pure effect of CAD geometry on the quality of the machined surface, linear interpolation of individual positions of CL (Cutter Location) points generated in CAM system is applied. Thus, the toolpath is represented by a discrete segmented linear curve (discrete curve) given by vertices $\mathbf{H}_i$, $i = 0, ..., m$, i.e. CL points. No further smoothing of the toolpath by arc or spline interpolation is considered here. To investigate, evaluate and assess the influence of various CAD geometries on quality of toolpath, two metrics of a discrete curve are introduced here: the length of segment (the distance of two consecutive CL points) and the discrete curvature in dependence on the length of the discrete curve.

The length of segment in dependence of the length of the discrete curve can be expressed by a pair of numbers (L_i, h_i), $i = 1, ..., m$, where h_i is the distance between two consecutive vertices, see fig. 2,

$$h_i = \|\mathbf{h}_i\| = \|\mathbf{H}_i - \mathbf{H_{i-1}}\| \tag{14}$$

and L_i is the length of the discrete curve from $\mathbf{H}_0$ to $\mathbf{H}_i$

$$L_i = \sum_{j=1}^{i} h_j. \tag{15}$$

The distance (14) represents the length of tangent vector $\mathbf{h}_i$ pointing from $\mathbf{H}_{i-1}$ to $\mathbf{H}_i$. Thus, the vertices of the discrete curve can be considered the points on this curve with arc-length parameterisation. The length of arc is given by (15). Consequently, with respect to (9), discrete curvature vector $\boldsymbol{\kappa}_i$ at the vertex $\mathbf{H}_i$ of a discrete curve can be introduced in terms of central finite

differences as follows

$$\boldsymbol{\kappa}_i = \frac{\hat{\mathbf{h}}_{i+1} - \hat{\mathbf{h}}_i}{\|\mathbf{h}_i^*\|}, \; i = 1, ..., m-1, \tag{16}$$

where $\hat{\mathbf{h}}_{i+1}$ and $\hat{\mathbf{h}}_i$ are unit vectors and $\mathbf{h}_i^*$ is the vector pointing from $\mathbf{H}_i^*$ to $\mathbf{H}_{i+1}^*$ given by

$$\mathbf{H}_i^* = \tfrac{1}{2}(\mathbf{H}_{i-1} + \mathbf{H}_i), \; \mathbf{H}_{i+1}^* = \tfrac{1}{2}(\mathbf{H}_i + \mathbf{H}_{i+1}). \tag{17}$$

The length κ_i of discrete curvature vector $\boldsymbol{\kappa}_i$

$$\kappa_i = \|\boldsymbol{\kappa}_i\| \tag{18}$$

is the discrete curvature. A suitable smoothness metric of a discrete curve can be expressed by a pair of numbers (L_i, κ_i), $i = 1, ..., m-1$. The sign of discrete curvature is given by orientation of the discrete curvature vector (16).

Note that curvature of a discrete curve has no general concept in literature [29] even if the curvature of discrete surfaces (meshes) is well defined [30]. Practical usage of (16) is very simple, does not require any evaluation of cyclometric functions [31], no unit or uniform distribution of discrete curve vertices is necessary [32] and can be applied on both discrete curves in two and three dimensions.

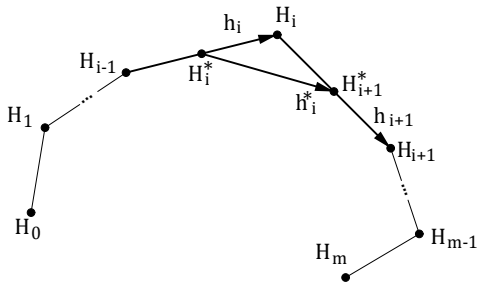


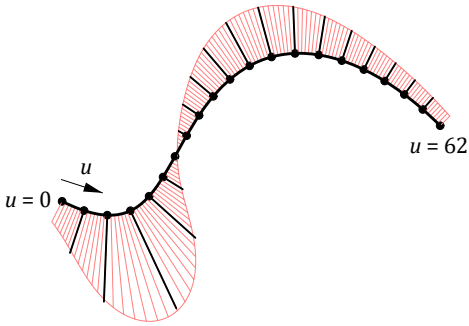
Fig. 2 Discrete curve

If the vertices $\mathbf{H}_i$ of the discrete curve are obtained by tessellation of a continuous curve, in the limit case by refining the tessellation, the length of the discrete curve (15) turns into the length of arc (7) of the continuous curve and the discrete curvature vector (16) turns into the curvature vector (6) of the continuous curve.

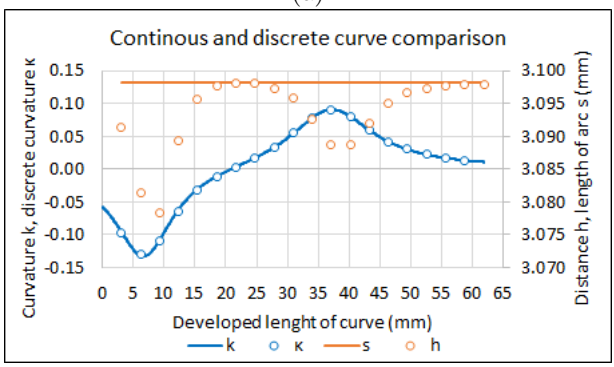
An example of a discrete linear curve given by tessellated points on the continuous curve is shown in fig. **??** (a), where a planar curve is shown for easy readability of the figure. The curvature graph of the continuous curve is

drawn in red, discrete curvature vectors at the vertices calculated according (16) are drawn in black.

To investigate the course of the length of segment and the discrete curvature of a discrete curve, it is better to draw a graph of both quantities depending on the developed length of the curve, see fig. 3 (b). From this graph, it is possible to compare the calculated values of discrete quantities h_i and κ_i with the corresponding values of continuous quantities $s(u)$ and $k(u)$ along the developed length of the curve drawn in fig. 3 (a).



(a)



(b)

Fig. 3 Continuous and discrete curve – (a) Curvature graph of continuous curve and discrete curvature at discrete points of the curve, (b) Curvature and length of arc of continuous curve and discrete curvature and length of segment at discrete points of the curve along the developed length of the curve

4 Motivation example

In this section, a motivation example of CAD model quality influence on the quality of the machined workpiece is presented. The artefact called Phobos, see fig. 4, is used as a test piece. The Phobos artefact was designed at the

Research Centre of Manufacturing Technology (Faculty of Mechanical Engineering, Czech Technical University in Prague) for research on the influence of technological parameters on the quality of the machined surface and production efficiency [33]. Here, geometrical properties of the Phobos artefact are described and their impact on the quality of the machined surface are examined.

The artefact in fig. 4 is a solid created by straight extrusion of a planar profile curve (fig. 5), which is closed by planar bases (fig. 6. The profile curve whose shape is similar to the shape of a compressor wheel blade profile intentionally consists of several segments (connecting points are marked with red numbers in the figures) in the following way: segments 0-1, 1-2, 5-6 and 6-0 are straight line segments, segments 3-4 and 4-5 are circular arcs and segment 2-3 is a uniform clamped B-spline curve of 10-th degree given by the drawn control polygon. Thus, the extrusion consists of planes, cylindrical surfaces of revolution and B-spline cylindrical surface in the given order. The CAD model of the Phobos artefact is depicted in fig. 6 together with the edges between the individual extruded surfaces and principal dimensions.

Fig. 4 Phobos – Manufactured artefact

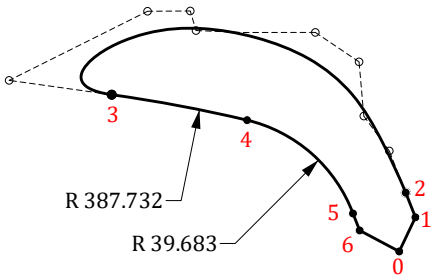


Fig. 5 Phobos – Profile curve

In the photo of the manufactured artefact shown in fig. 4, undesired grooves (marked by red numbers) are clearly visible. These grooves are located exactly in places 3, 4 and 5, where the individual extruded profile segments are connected.

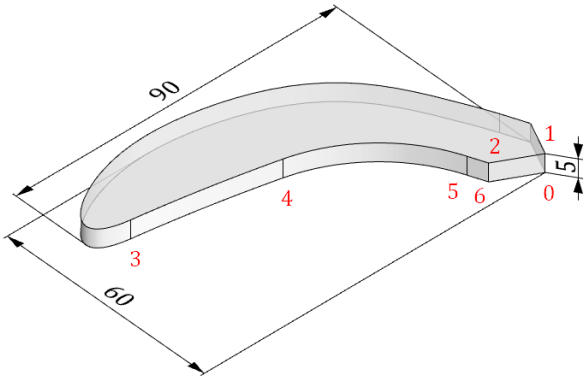


Fig. 6 Phobos – CAD model

To understand why grooves are formed and how to avoid them in the first phase, i.e. when designing a CAD model of the manufactured workpiece, it is necessary to examine the geometry of the CAD model of the Phobos artefact and derive the influence of geometric properties on the kinematic properties of the cutting tool.

4.1 Quality of CL points assessment

If the possibility of setting parameters such as toolpath strategies and tool position tolerances is omitted, then the distribution of CL points depends only on the quality of the CAD model. Thus, in this section, CL points for 5-axis ball-end (radius 5 mm) milling generated in CAM system based on three CAD models of the Phobos artefact differing in their geometrical properties are analysed. Cutting conditions for all three simulation cases were the same.

CL points analysis by means of colour visualisation of the length of toolpath segments, currently the only way to estimate the quality of CL points, is mentioned first. After that, a new characteristics based the relation between the discrete curvature of a toolpath and the manufactured profile curvature is introduced and demonstrated.

4.1.1 Length of toolpath segments

Usually, only the length of toolpath segments can be visualised in CAM system in form of coloured positions of CL points along the developed profile, see fig. 7. The colour of points corresponds to the distance between two consecutive CL points. The length of segments affect the velocity of the cutting tool which must move more slowly in places where the length of segments is shorter. In the contrary, in places where the points are farther apart, the cutting tool can afford higher velocity. It follows that the rate of change of CL points distance affects the smoothness of the cutting tool movement and thus the quality of the machined surface. The quality of CL points is estimated according to the

smoothness in colour transition along the manufactured surface, which may not always lead to unambiguous conclusions.

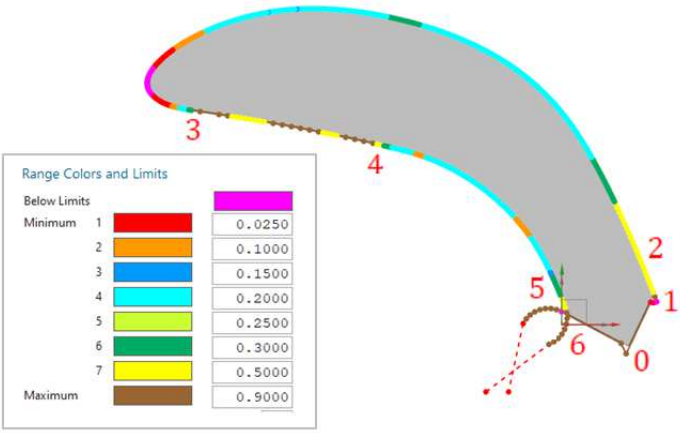


Fig. 7 Colour visualisation of the distance of CL points

4.1.2 Toolpath discrete curvature

Upon closer examination of geometrical properties of the profile of the Phobos artefact visualised in the curvature graph in fig. 8, it is obvious that the curvature of the profile is discontinuous at points 2, 3, 4 and 5 where the individual extruded surfaces meet. According to continuity analysis shown in fig. 1, these surfaces are connected with G^1 continuity only. Obviously, such a poor order of continuity leads to sharp changes in position and distance of CL points and finally to undesired grooves on the machined surface. The higher the curvature of the machined profile, the shorter the distance between CL points. Due to the dynamics properties of the machine tool, the cutting tool that is controlled by an interpolated path containing areas with sharply compressed CL points must decelerate in front of these areas. This deceleration causes the above-mentioned grooves on the workpiece, see fig. 4.

It can be reasonably assumed that a higher order of continuity will lead to a more even movement of the cutting tool. To confirm this assumption, not only the original CAD model of the Phobos artefact was considered, but also the CAD models with a single cubic and quartic B-spline curves were analysed. Both CAD models were created so that the deviation of the shape of the cubic and quartic profiles from the original one was up to 0.1 mm, see *Profile curve deviation* characteristic in tab. 1. Control points and curvature graphs of cubic and quartic profiles are drawn in figs. 9 and 10.

The influence of CAD model continuity on the quality and efficiency of the machining process was tested in the following steps, applied on all three considered profiles of the Phobos artefact (original, cubic and quartic).

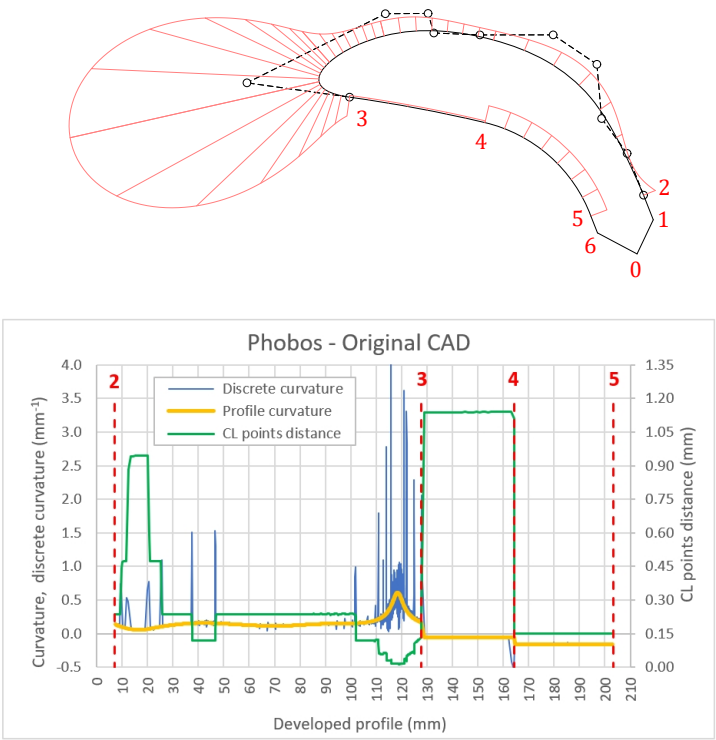


Fig. 8 Phobos – Influence of G^1 continuous original CAD profile on the quality of CL points

Procedure for testing the influence of CAD model continuity on quality of CL points

1. Generation of CL points in CAM system in a helix toolpath with lead about 0.300 mm.
2. Evaluation of geometrical form error, i.e. the range of deviations of the machined profile from the theoretical profile given by the original CAD model, see tab. 1. In this step, no dynamic properties of the machining process are considered.
3. Separation of the set of vertices $\mathbf{H}_i$, $i = 0, ..., m$ for the definition of discrete curve. This set is given by individual positions of the tool tip belonging to one thread of the toolpath from $z = 2.333$ mm to $z = 2.500$ mm.
4. Calculation of pairs of numbers (L_i, h_i), $i = 1, ..., m$ given by (14) and (15) to measure the length of segments of the toolpath. Extreme values of h_i (CL points distances) determine the range of the length of segments, see tab. 1.
5. Calculation of pairs of numbers (L_i, κ_i), $i = 1, ..., m$ given by (14) and (18) to measure the discrete curvature of the toolpath. Extreme values of κ_i determine the range of the toolpath discrete curvature, see tab. 1.
6. Determination of the first profile curvature $k(u)$, i.e. the length of profile curvature vector $\mathbf{k}(u)$ given by (6).

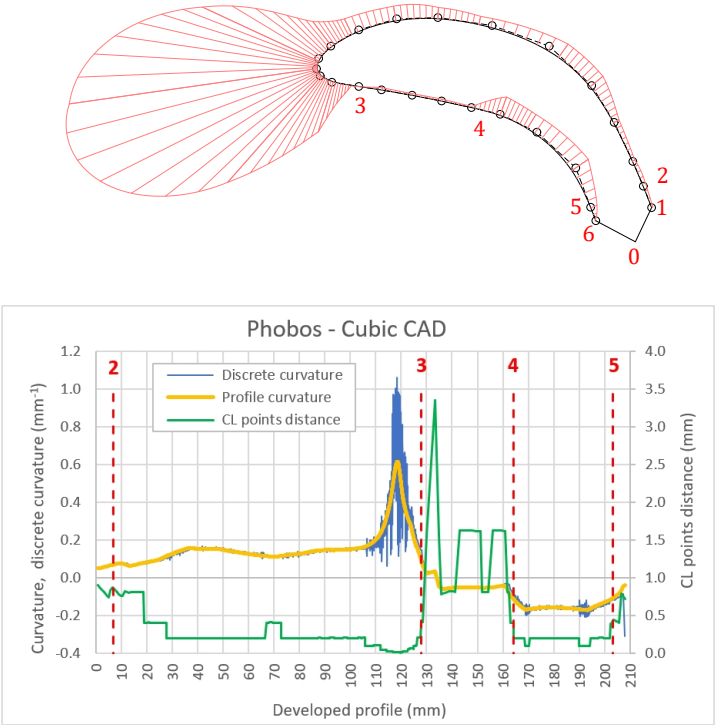


Fig. 9 Phobos – Influence of G^2 continuous cubic CAD profile on the quality of CL points

7. Drawing a graph of the profile first curvature $k(u)$, discrete toolpath curvature κ_i and the distance h_i of CL points along the developed profile. These graphs are drawn in figs. 8 – 10. Due to extreme values of the toolpath discrete curvature, signed square roots of the curvatures (6) and (16) are drawn in the graphs.

When comparing the graphs in figs. 8 – 10 and the detail of the area between points 3 and 4 in fig. 11, it is obvious that the quality of the discrete toolpath strongly depends on the curvature of the manufactured profile. Sharp changes in the length of segments and high oscillations whenever the original CAD profile curvature was discontinuous is clearly visible in the graphs.

The cubic CAD profile provides C^2 continuity and shows better results. However, the sharp changes in the length of segments and the discrete curvature are still present. Specifically, the discrete curvature has the widest range from all the considered CAD models, see tab. 1.

The quartic CAD profile ensures C^3 continuity along the whole profile and, therefore, the graph of the length of segments as well as discrete curvature shows no sharp changes and, especially in neighbourhood of points 3 and 4, the undesirable oscillations are strongly limited.

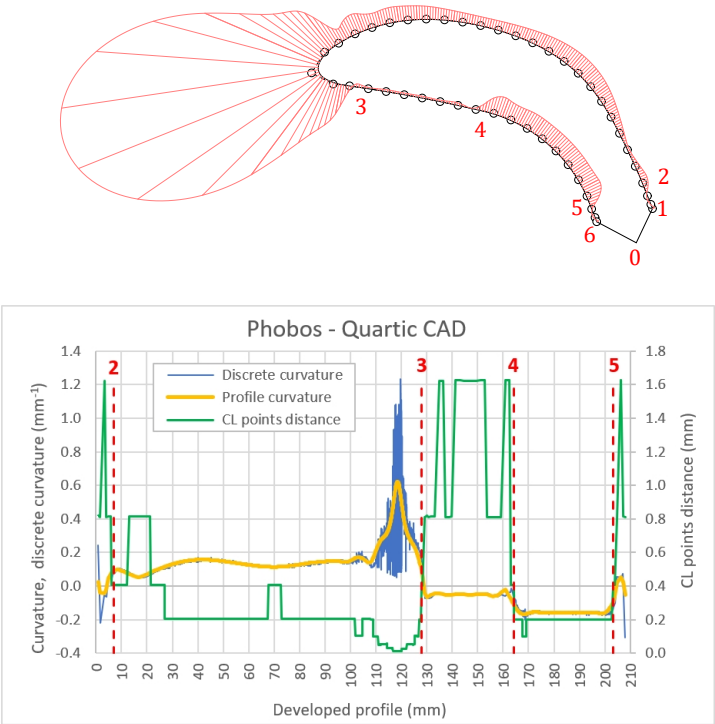


Fig. 10 Phobos – Influence of G^3 continuous quartic CAD profile on quality of CL points

Table 1 Phobos – Characteristics

Characteristic	Original CAD	Cubic CAD	Quartic CAD
Profile deviation (mm)	0.000	0.077	0.047
Geometrical form error (mm)	[-0.082,0.368]	[-0.035,0.078]	[-0.031,0.047]
Length of segment (mm)	[0.012,1.141]	[0.010,3.345]	[0.013,1.626]
Profile curvature (mm^{-1})	[-0.025,0.373]	[-0.029,0.379]	[-0.028,0.387]
Toolpath curvature (mm^{-1})	[-0.281,16.223]	[-0.096,1.123]	[-0.307,1.235]

To conclude the achieved results from the above given analysis, high values of the first profile curvature are not a serious problem from the point of view of CL points generation. The problem consists in sharp changes or overshoots of the first curvature. A shorter distance of CL points causes a lower tool velocity. If the distance of CL points changes sharply, the tool velocity changes sharply, which causes a machined surface of a worse quality. Therefore, the ideal geometry of the manufactured profile consists of a G^3 continuous curve of at least 4-th degree with a gradual change of the first curvature [34].

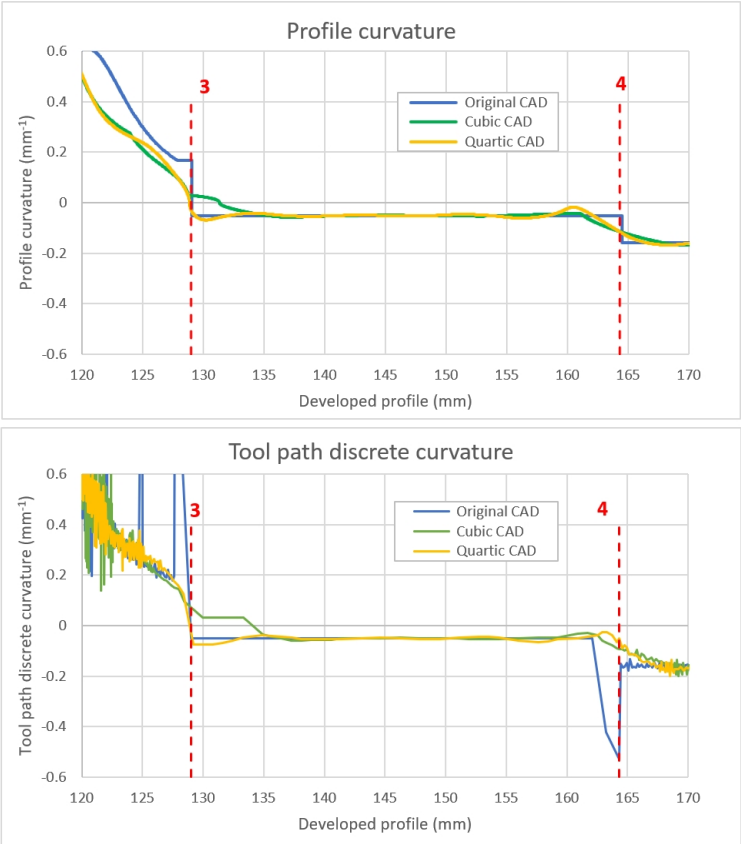


Fig. 11 Phobos – Graph of profile (top) and discrete toolpath curvature (bottom) along the developed original, cubic and quartic CAD profiles

5 Experimental verification

Experimental verification of the above stated influence of CAD model continuity on the quality and efficiency of 5-axis CNC machining is documented in this section. The experiment was performed on a Blisk profile artefact that was designed and manufactured for the purposes of this paper. The CAD model of the Blisk profile artefact has two functional parts — Original CAD profile and Quartic CAD profile, both created by extrusion of a planar definition curve, see fig. 12, where the CAD model together with basic dimensions are drawn on the left. On the right, the photograph of the manufactured artefact is shown.

In this paper, the profile manufactured according to the Original CAD profile is hereinafter referred to as the Original profile and the profile manufactured according to the Quartic CAD profile is referred to as the Quartic profile.

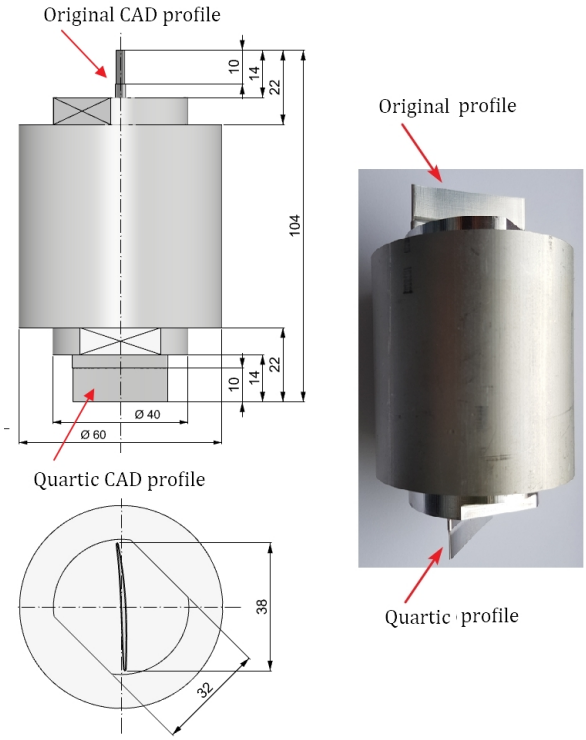


Fig. 12 Blisk profile artefact – Technical drawing (left) and photo of the manufactured artefact (right)

The definition curve of the Original CAD profile is given by a suitably placed planar section of a single Blisk blade surface drawn in fig. 13. According to the curvature graphs shown in fig. 13, the Blisk blade surface is G^0 continuous only. Therefore, the suitable position of the planar section for the definition of the Original CAD profile was chosen at the place with the highest discontinuity in curvature, see the resulting curve in fig. 15 (a).

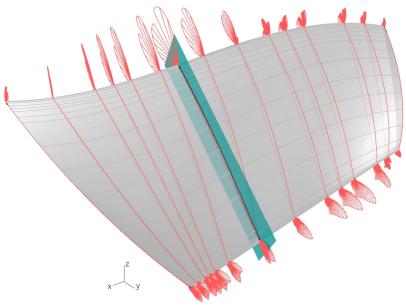


Fig. 13 Single Blisk blade and curvature analysis

5.1 Experiment settings

NC code is generated using the Siemens NX CAM software for the HEIDENHAIN TNC640 control system which is equipped with the MCU 700 VT-5X multifunctional CNC machine tool from KOVOSVIT MAS Machine Tools, a.s., see fig. 14.

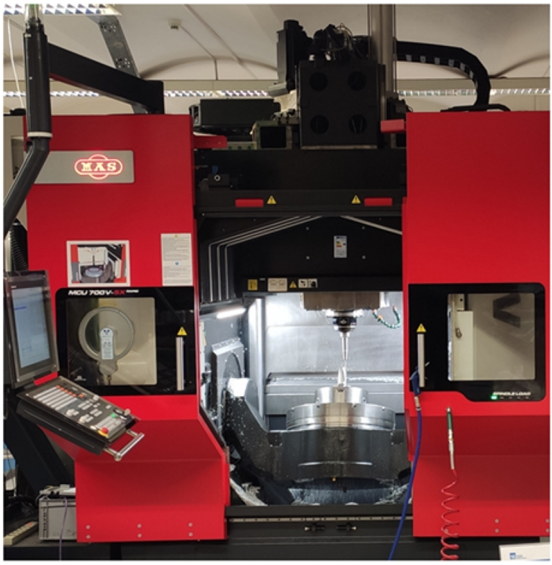
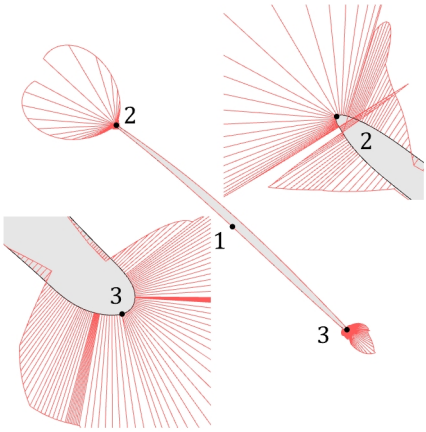


Fig. 14 Machine tool MCU 700V[T]-5X with Heidenhain TNC640

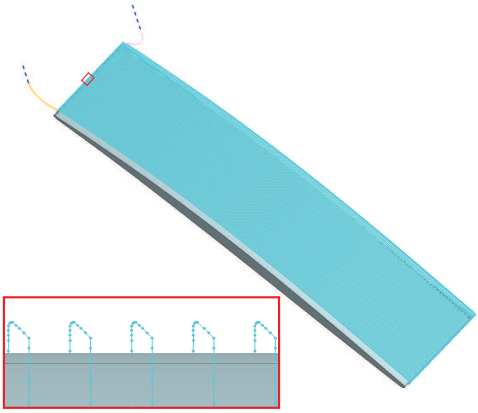
The Blisk profile artefact was manufactured from aluminum alloy EN AW 7075 T6 and machined with a ball-end cutter with the required tolerance zone [0.00, 0.09] mm. Machining strategy was helix from up to down. Stock of material for finishing was 0.15 mm. Parameters of the experiment settings are given in tab. 2.

Table 2 Experiment setting

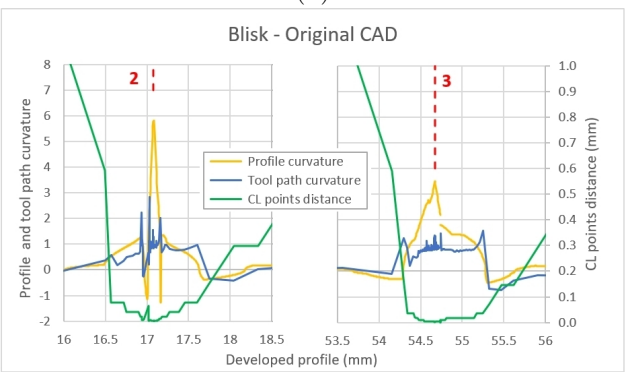
Parameter	Value
Coolant	ON
Lead angle	0°
Tilt angle	75°
Spindle speed	12 000 RPM
Feedrate	2 000 mm min^{-1}
Tool radius	5 mm
Tooth number	4
Stock material for finishing	0.15 mm
Stepover	0.17 mm
Material	EN AW 7075 T6



(a)

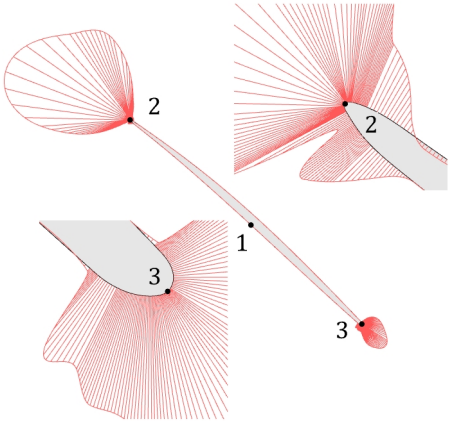


(b)

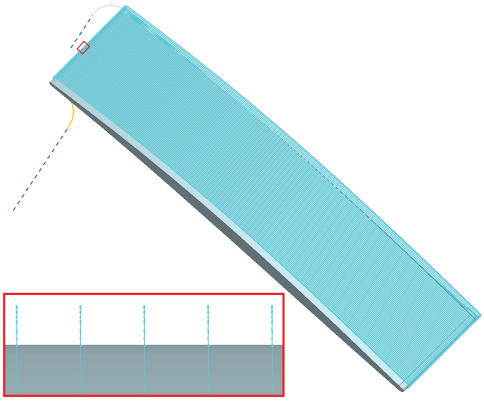


(c)

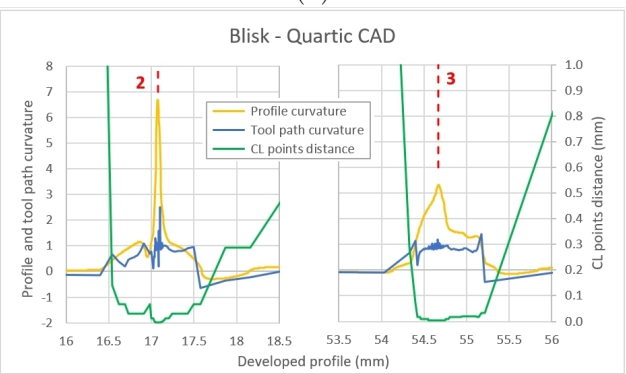
Fig. 15 Blisk profile artefact (Original CAD profile) – (a) Curvature graph, (b) CL points, (c) Profile and toolpath discrete curvature



(a)



(b)



(c)

Fig. 16 Blisk profile artefact (Quartic CAD profile) – (a) Curvature graph, (b) CL points, (c) Profile and toolpath discrete curvature

5.2 CAD model of Blisk profile artefact analysis

The curvature graph of the Original CAD profile is depicted in fig. 15 (a). In the detail, sharp changes of curvature at points 2 and 3 are obvious. The

definition curve for the Quartic CAD profile is given by replacing the original profile curve with a quartic B-spline curve so that the geometrical form error (0.006 mm) of the quartic B-spline curve with respect to the original profile curve is negligible with respect to the tolerance zone [0.00, 0.09] mm. The definition curve for the Quartic CAD profile is G^3 continuity with minimal curvature overshoots and a gradual change in the curvature, see fig. 16 (a).

On both the Original and Quartic CAD profiles, CL points have been generated, see fig. 15 (b) and fig. 16 (b), respectively. CL points generated on the Original CAD profile are unevenly distributed and show sharp changes in the direction of the toolpath, as is visible in the detail in fig. 15 (b). A significant improvement in the shape of the toolpath for the Quartic CAD profile is shown in fig. 16 (b).

The quality analysis of both toolpaths has been performed according to the procedure described in section 4. The resulting graphs for the Original and Quartic profiles are drawn in fig. 15 (c) and fig. 16 (c), respectively. In these graphs, only the details of the neighbourhood of points 2 and 3, where the first curvature of the Original CAD profile shows the highest changes, are depicted.

By comparing all the monitored characteristics of the Original and Quartic CAD profiles, it is evident that the overshoots of the first curvature at location 2 and the discontinuity at location 3 were removed. The oscillation of the toolpath curvature of the Quartic CAD profile compared to the Original CAD profile has been reduced and the distance of the CL points is also smoother.

5.3 Inspection of the artefact

In this section, the influence of CAD model continuity on the quality and efficiency of 5-axis ball-end milling is summarised. This influence is assessed by means of visual inspection, coordinate profile deviations measurement and surface roughness and machining time measurement of both manufactured profiles – Original and Quartic, see fig. 12 right.

5.3.1 Visual inspection

The visual comparison in fig. 17 of both manufactured profiles shows an undesirable groove on the Original profile (left) caused by a significant deceleration of the milling tool in front of point 2, where the curvature changes abruptly. On the Quartic profile (right), there are no unwanted traces of the milling tool visible. Thus, the influence of the higher order continuity of the CAD model on the quality of the machined surface is visible to the naked eye.

5.3.2 Shape accuracy inspection

Original and Quartic profiles were measured in the Czech Metrology Institute on the National standard of geometrical dimensions of 3D objects – coordinate measuring machine ZEISS XENOS [35], see fig. 18.

In fig. 19, the graph of the deviations of the manufactured profiles from the nominal CAD references is depicted. The limits [0.000, 0.090] mm of the

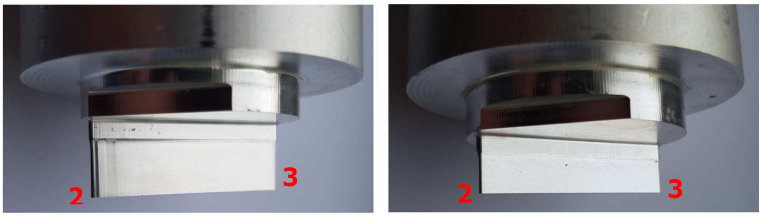


Fig. 17 Blisk profile artefact – Original profile (left) and Quartic profile (right)

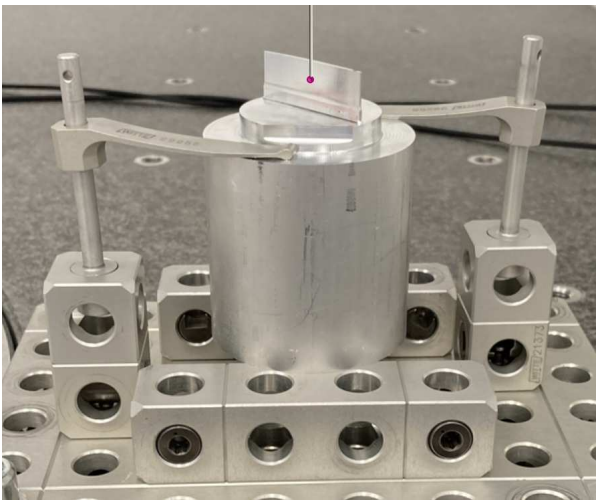


Fig. 18 Blisk profile artefact – Measurement on ZEISS XENOS

tolerance zone and maximum and minimum deviations of the measured profiles from the nominal shape are indicated in this graph, too. The range of deviations measured along the Original profile is [0.027, 0.103] mm and along the Quartic profile [0.011, 0.070] mm. The deviations of the Original profile are out of the tolerance zone and the form error 0.076 mm is bigger than form error 0.059 mm of the Quartic profile whose deviations are in the tolerance zone. The maximum deviation was reduced from 0.103 mm to 0.076 mm, which represents an improvement of manufacturing accuracy of 31.96 %.

5.3.3 Surface roughness inspection

The roughness of the artefact surface was measured using a roughness tester from Mahr LD130 using a measuring sensor LP C 25-15-2-90, see fig. 20. The evaluation was performed according to the ISO 4287 standard, using a Gaussian filter ISO 16610-21 with limiting wavelength LC of 0.8 mm. Roughness measurement was performed in a direction perpendicular to the tool feed direction (parallel to the tool axis) and in the same direction as the tool feed direction. The measurement was repeated five times for each direction, see tab. 3. The measurement result shows the same roughness for both machining

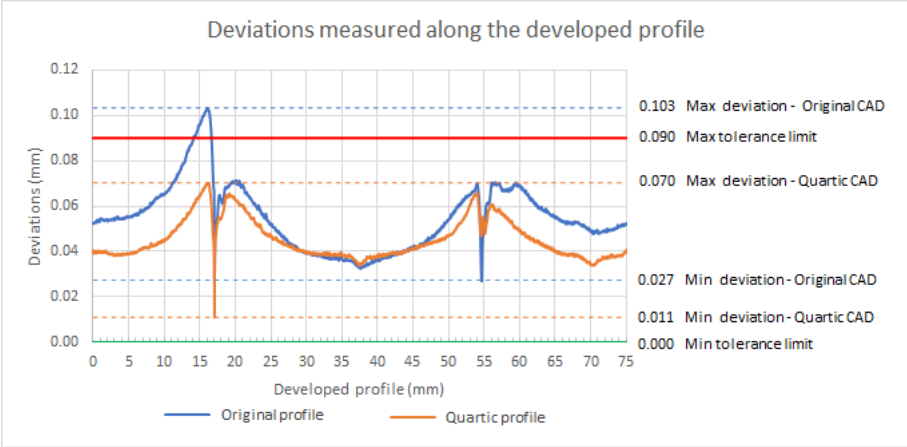


Fig. 19 Comparison of deviations measured along the profiles manufactured according to the Original and Quartic CAD profiles

cases – Original and Quartic profile. Modification of the CAD model does not affect the surface roughness.

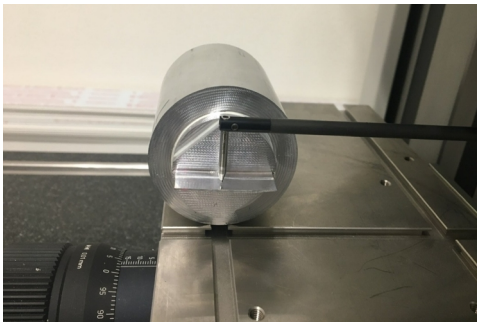


Fig. 20 Roughness measurement – Mahr LD130

5.3.4 Machining time

Machining time was measured during machining by the Heidenhain TNC640 control system. The number of passes, CAM tolerances and programmed feedrate were the same for both the Original and Quartic profiles. Machining time for the Original profile was 11:51 minutes and for the Quartic profile it was 11:25 minutes. Modification of the CAD model has a positive effect on shortening the production time, in this case by 3.9 %.

5.4 Evaluation of experimental verification

The results of all measurements are summarised in tab. 5. It can be stated that increasing the continuity of the CAD model from G^0 to G^3 while maintaining

Table 3 Perpendicular roughness measurement result (all dimensions are in micrometres)

Measurement	Original profile			Quartic profile		
	Rz	Rsm	Ra	Rz	Rsm	Ra
1	1.30	110	0.26	1.29	117	0.22
2	1.22	126	0.24	1.34	100	0.24
3	1.24	90	0.23	1.17	78	0.22
4	1.30	118	0.26	1.37	129	0.24
5	1.16	92	0.23	1.17	93	0.21
Average	1.24	107	0.24	1.27	103	0.23

Table 4 Feed roughness measurement result (all dimensions are in micrometres)

Measurement	Original profile			Quartic profile		
	Rz	Rsm	Ra	Rz	Rsm	Ra
1	1.04	214	0.21	1.19	197	0.22
2	1.07	143	0.20	1.20	233	0.25
3	1.06	204	0.22	1.10	193	0.20
4	1.20	219	0.22	1.02	204	0.21
5	0.96	220	0.21	1.04	259	0.22
Average	1.06	200	0.21	1.11	217	0.22

the shape accuracy well below the tolerance zone results in an improvement in both the quality and efficiency of 5-axis ball-end machining. A higher quality CAD model therefore results in higher accuracy of the machined surface by reducing deviations and at the same time increasing machining productivity without affecting the surface roughness.

Table 5 Blisk profile artefact – Experiment verification

Characteristics	Original CAD	Quartic CAD
CAD profile deviation (mm)	0.000	0.006
CAD profile curvature (mm^{-1})	[-1.618, 33.411]	[-0.086, 44.737]
Toolpath curvature (mm^{-1})	[-0.540, 8.081]	[-0.422, 6.225]
Length of segments (mm)	[0.000, 2.349]	[0.001, 2.352]
Manufacturing deviation (mm)	[0.027, 0.103]	[0.011, 0.070]
Perpendicular roughness Ra (μm)	0.24	0.23
Feed roughness Ra (μm)	0.21	0.21
Machining time (min)	11:51	11:25

6 Conclusion

In the paper, a significant influence of CAD model continuity on the quality and efficiency of the surface machined by 5-axis ball-end milling is examined and demonstrated. It is shown that the quality of the toolpaths derived from the CAD model is most affected by the curvature of the surface itself. High values of the first profile curvature do not represent a serious problem if the change of the curvature is gradual and does not oscillate sharply. Jump changes occur in

surfaces of the third and lower degree, in which there is a non-smooth or even discontinuous course of the curvature graph. Oscillating curvature causes the oscillation of the distance of CL points, which negatively affects the smooth movement of the tool.

A fast change of the first curvature also means the fast change density of CL points and fast change of the tool movement. For the purpose of CNC machining, the aim therefore is to use curves (surfaces) of at least fourth degree.

Based on arc-length parameterisation and central finite differences, a method of a discrete curvature evaluation of a discrete linear curve is introduced in this paper. If the discrete curvature evaluation is applied on CL points, it is possible to predict the places which would cause an undesired reduction in the quality and accuracy of the machined surface during machining. Compared to the discrete curvature calculation methods used so far, the method presented in this paper does not require any evaluation of cyclometric functions. This approach does not even require unit or uniform arrangement of the discrete points. Additionally, it can be applied to both planar and spatial discrete curves.

Based on experimental verification, where Blisk artefact was machined from aluminium alloy on 5-axis machine tool, the benefit and influence of CAD model continuity on production was proved. It has been verified that by increasing the continuity of the CAD model, it is possible to increase the accuracy of the machined surface by reducing deviations and at the same time increasing machining productivity without affecting the surface roughness.

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Author Contributions

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